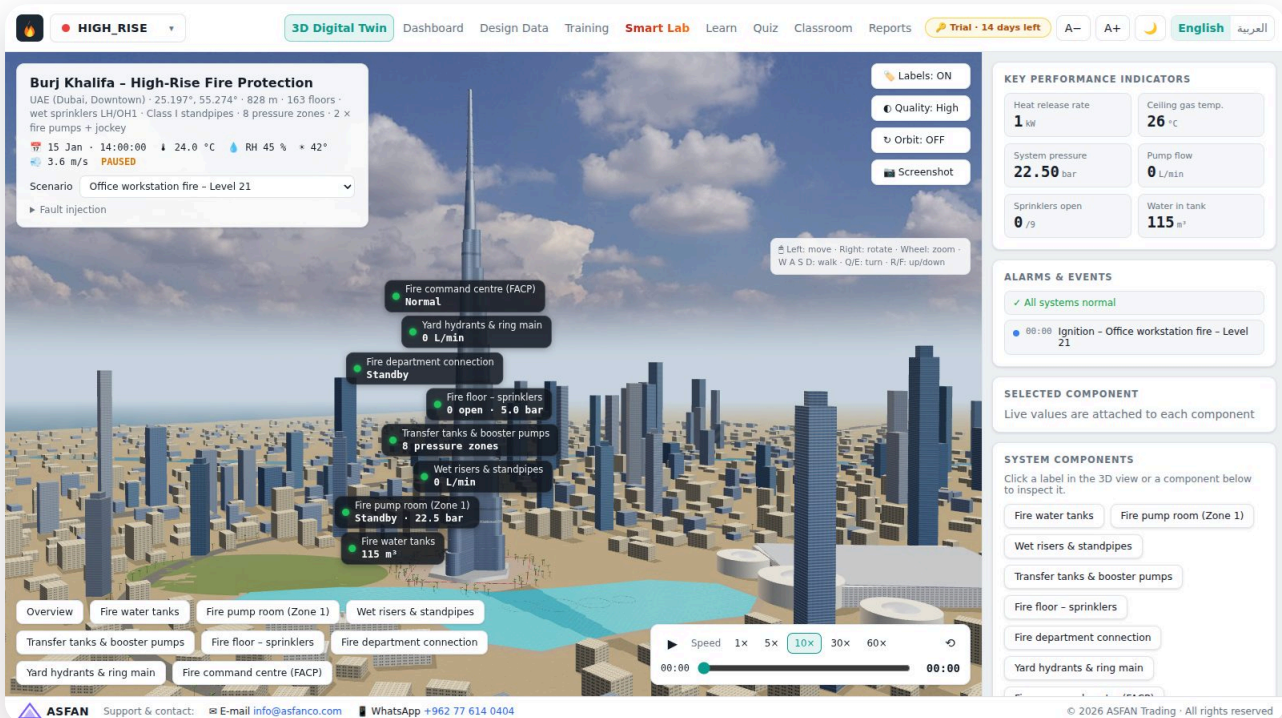




# Fire Protection Digital Twin

User Manual · Educational fire-fighting & smart fire-alarm simulator

Version 1.1 · Windows 10 / 11 (64-bit) · English



ASFAN Trading

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**Fire Protection Digital Twin** is an educational real-time simulator of fire-fighting and fire-alarm systems for universities, training centres and companies. Real sites (Burj Khalifa, Aqaba port, Amman data centre, oil tank farm, airport hangar) are modelled in 3D with their fire-protection systems designed to NFPA standards. A physics engine simulates fire growth, detection, sprinkler operation, pumps, water supply, clean agents and foam — and every value is shown live on the 3D model.

## What you can do

- Watch complete fire scenarios in 3D, with live pressures, flows, temperatures and alarms on every component.
- Inject faults (power failure, closed valve, jockey pump failure, leaks...) and diagnose them.
- Edit the design data and run the simulation on your own design, with NFPA compliance checks.
- Practise real procedures in ten interactive 3D training scenes, scored step by step.
- Learn smart (addressable) fire-alarm systems in the Smart Systems Lab: panel, cause & effect, BMS integration, commissioning, maintenance and incident command.
- Study lessons, take quizzes, run classroom challenges, print reports and certificates.

### Educational use

Building data are public information; the fire systems are NFPA-based reference designs for teaching. Do not use results for real projects without verification by a qualified engineer.

## 2 System requirements

---

|                  | Minimum                                                         | Recommended                        |
|------------------|-----------------------------------------------------------------|------------------------------------|
| Operating system | Windows 10 64-bit                                               | Windows 11 64-bit                  |
| Processor        | Intel Core i5 (8th gen) / Ryzen 5                               | Core i7 / Ryzen 7                  |
| Memory           | 8 GB RAM                                                        | 16 GB RAM                          |
| Graphics         | Integrated GPU with DirectX 11 / WebGL 2                        | Dedicated GPU (NVIDIA / AMD), 2 GB |
| Screen           | 1366 × 768                                                      | 1920 × 1080 or larger              |
| Disk space       | about 500 MB                                                    |                                    |
| Internet         | Not required (only for the optional YouTube source-video links) |                                    |

## 3 Installing the program

You receive one of two files from ASFAN:

| File                                         | Use                                                                                                                       |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| FireProtectionDigitalTwin-Setup-1.1.0.exe    | <b>Installer (recommended).</b> Installs into Program Files, creates desktop and Start-menu shortcuts and an uninstaller. |
| FireProtectionDigitalTwin-Portable-1.1.0.exe | <b>Portable.</b> Runs directly without installation — e.g. from a USB stick or a lab PC without admin rights.             |

### Installation steps

- 1 Double-click `FireProtectionDigitalTwin-Setup-1.1.0.exe`.
- 2 Windows may show a blue “**Windows protected your PC**” window (Microsoft SmartScreen). Click **More info**, then **Run anyway**. This appears because the program is new and is not yet signed with a paid code-signing certificate — it is safe to continue.
- 3 Choose the installer language (English / العربية) and accept the license agreement.
- 4 Choose **for all users** or **only for me**, and the installation folder (the default is fine).
- 5 Click **Install**, then **Finish**. The program starts automatically; a shortcut **Fire Protection Digital Twin** is placed on the desktop.

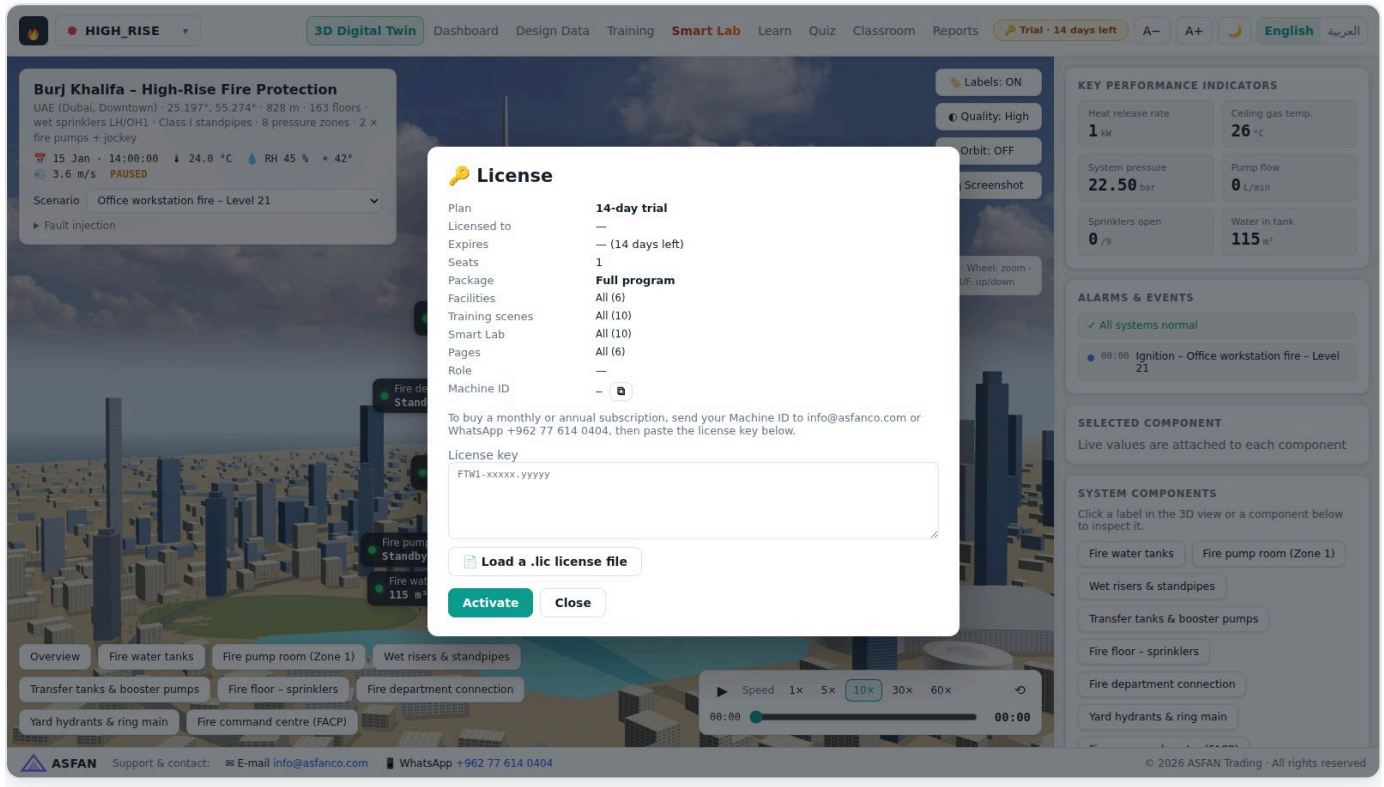
#### Antivirus

Some antivirus programs may scan the new file for a few seconds or ask for confirmation the first time. Allow the file — it is the official ASFAN installer. If a school network blocks unknown programs, ask the IT administrator to allow the installer or use the portable version.

### Updating and uninstalling

- **Update:** run the new Setup file — it replaces the old version. Your license, settings and results are kept.
- **Uninstall:** Windows Settings → Apps → Fire Protection Digital Twin → Uninstall.

On the first start the program runs as a **14-day full trial**. The license badge at the top (🔑) shows the remaining days.



The license window: plan, customer, expiry, package contents and the Machine ID

## Buying and activating a subscription

- 1 Click the 🔑 badge at the top of the window (or **Help** → **License...**).
- 2 Copy your **Machine ID** with the 📄 button and send it with your order to ASFAN (e-mail or WhatsApp) — only needed if your license should be bound to this computer.
- 3 ASFAN sends you a license file `.lic` (or a key starting with `FTW1-`).
- 4 In the license window click **Load a .lic license file** (or paste the key) and press **Activate**. The badge changes to **Licensed · Monthly / Annual**.

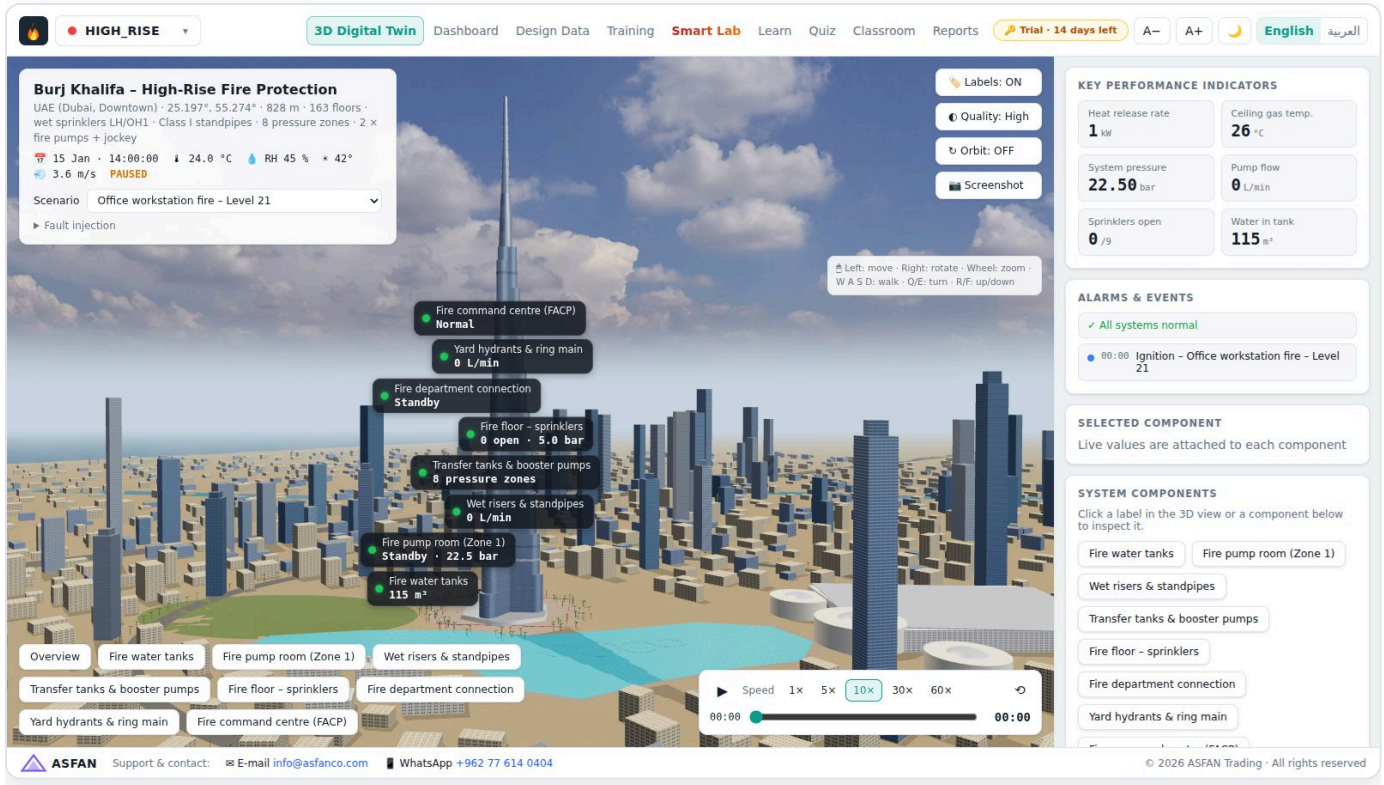
## Packages

A license can include the **full program** or only a **package**, for example *Sprinklers & pumps*, *Smart fire alarm*, *Special hazards* or *Basic awareness*, or a custom selection. The license window lists exactly which facilities, training scenes, Smart Lab modules and pages are included. Everything else is shown with a 🔒 lock; clicking it opens the license window so you can upgrade.

| Situation                    | What happens                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| Subscription expires         | The program asks for a new key. Your data and results are kept.                                        |
| New computer                 | Unbound licenses can be activated on the new PC; bound licenses need a new key for the new Machine ID. |
| Supervisor (teacher) license | Sees, exports and clears the results of all students and the instructor dashboard.                     |

### **Do not change the computer date**

Setting the clock back is detected and blocks the program until the date is corrected.



Main screen: top bar, 3D view with live labels, information card, KPIs, alarms and components

## Top bar

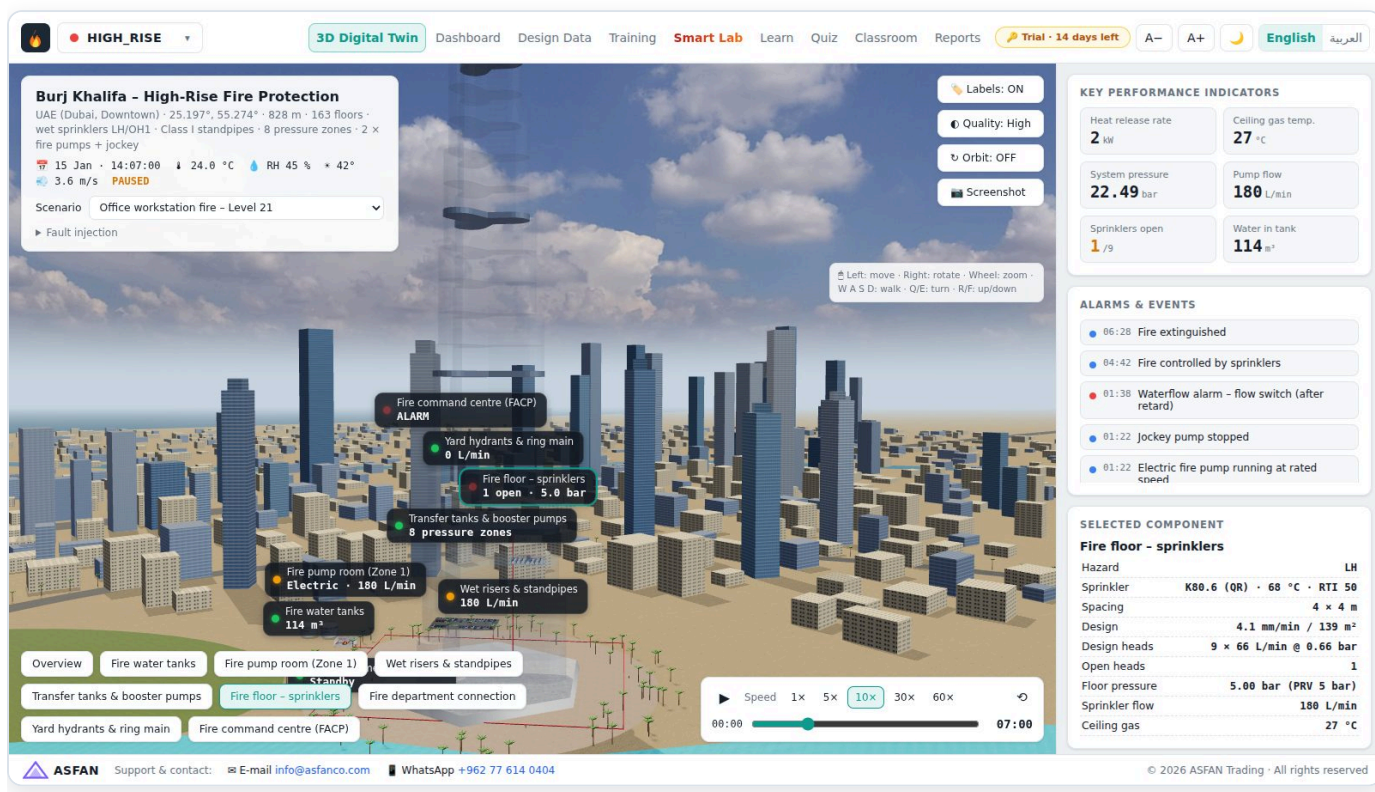
| Element           | Function                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------|
| Facility selector | Choose one of the reference facilities or <b>Create facility manually</b> (your own building).         |
| Tabs              | 3D Digital Twin · Dashboard · Design Data · Training · Smart Lab · Learn · Quiz · Classroom · Reports. |
| 🔑 badge           | License status and activation.                                                                         |
| A- / A+           | Smaller / larger text.                                                                                 |
| 🌙 / ☀️            | Dark / light theme.                                                                                    |
| ?                 | Opens this user manual.                                                                                |
| English / العربية | Switch the whole program language (Arabic is right-to-left).                                           |

## Moving in 3D

| Action                  | Mouse / keyboard  |
|-------------------------|-------------------|
| Move (pan)              | Left button drag  |
| Rotate                  | Right button drag |
| Zoom                    | Mouse wheel       |
| Walk                    | W A S D           |
| Turn                    | Q / E             |
| Up / down               | R / F             |
| Play / pause simulation | Space             |

Buttons on the right of the 3D view: **Labels** on/off, **Quality** high/low (use *low* on slow laptops), **Orbit** auto-rotation, and **Screenshot** (also **Ctrl** + **Shift** + **S** ).

## 6 3D Digital Twin

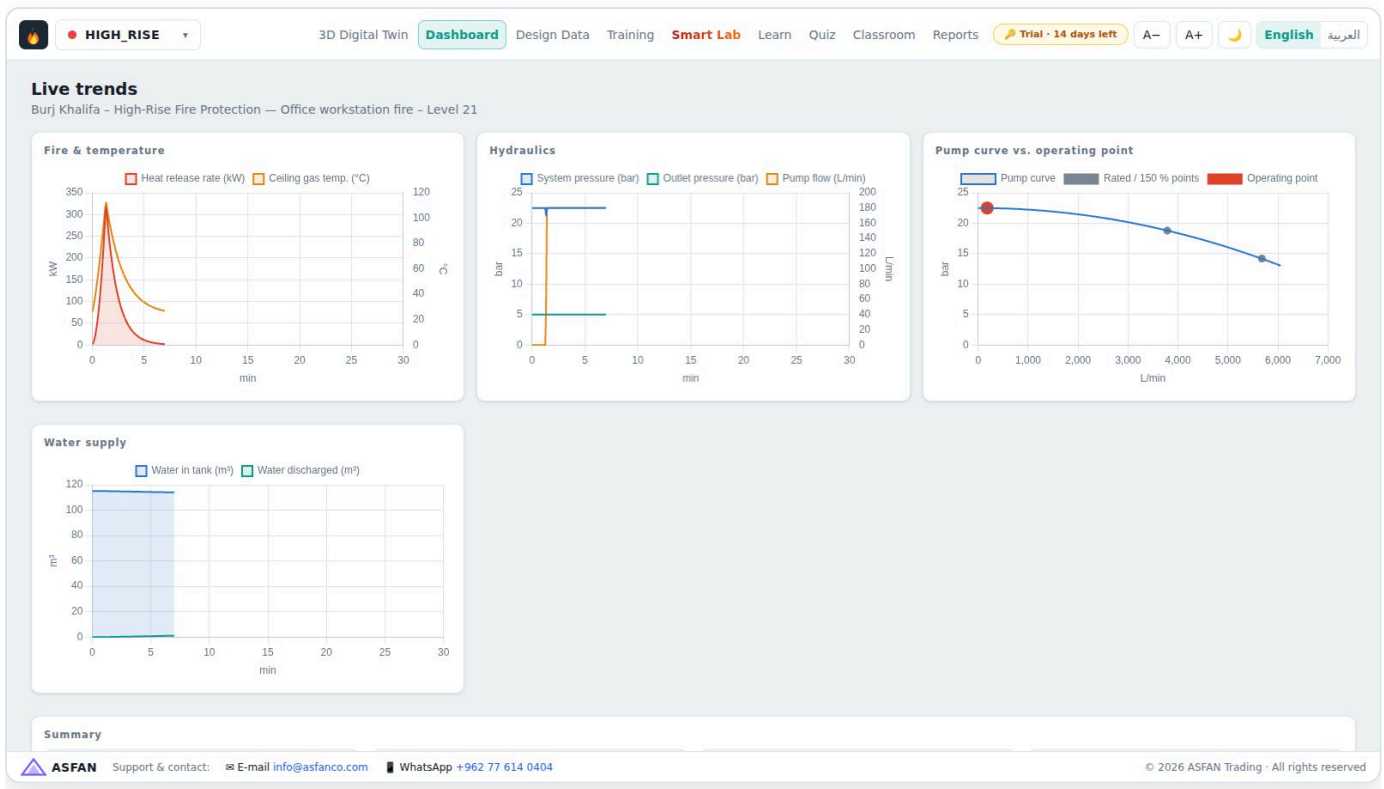


A fire scenario running: sprinklers operating, live pressures and alarms

- 1 Select a **facility** and a **scenario** (e.g. fire on a given floor).
- 2 Press ► to play. Use the speed buttons (1×–60×) and drag the time slider to jump.
- 3 Click any label or component to fly to it; its live values appear in **Selected component**.
- 4 Watch **KPIs** (heat release, gas temperature, pressure, flow, sprinklers open...) and the **Alarms & events** log.
- 5 Open **Fault injection** to add failures (power failure, closed valve, leak, pump failure...) and see how the system reacts.

### Create facility manually

Choose it from the facility selector to build your own building (city, occupancy, floors, dimensions, sprinkler K-factor, temperature rating, response type). The design and simulation are generated automatically.



Dashboard: fire, hydraulics, pump curve with moving operating point and water supply

Live trend charts for the running scenario: heat release and gas temperature, system pressure and flow, the fire-pump curve with the moving operating point, and tank/agent/foam supply. Charts follow the simulation time.

**Design basis & calculations**  
Burj Khalifa – High-Rise Fire Protection — Office workstation fire – Level 21

**Edit design data - your own values**  
Change any value, then run the simulation on your data. Changed fields are highlighted; values are saved per facility and scenario.

**Sprinkler system (NFPA 13)**

|                        |                                         |                                     |                                                       |                               |                               |
|------------------------|-----------------------------------------|-------------------------------------|-------------------------------------------------------|-------------------------------|-------------------------------|
| Occupancy hazard<br>LH | Sprinkler K-factor (L/min/vbar)<br>80.6 | Sprinkler spacing (square) (m)<br>4 | Response time index RTI ((m·s) <sup>1/2</sup> )<br>50 | Temperature rating (°C)<br>68 | Hose allowance (L/min)<br>379 |
|------------------------|-----------------------------------------|-------------------------------------|-------------------------------------------------------|-------------------------------|-------------------------------|

Water supply duration (min)  
30

**Piping & building**

|                            |                             |                                           |                                         |                         |                       |
|----------------------------|-----------------------------|-------------------------------------------|-----------------------------------------|-------------------------|-----------------------|
| Main pipe size (mm)<br>150 | Main pipe length (m)<br>140 | Fire floor elevation above pump (m)<br>88 | Floor PRV setting (0 = none) (bar)<br>5 | Ceiling height (m)<br>3 | Room length (m)<br>24 |
| Room width (m)<br>16       | Fire position x (m)<br>13   | Fire position y (m)<br>7                  |                                         |                         |                       |

**Fire scenario**

|                               |                                     |                                |
|-------------------------------|-------------------------------------|--------------------------------|
| Fire growth rate (t²)<br>fast | Peak heat release rate (kW)<br>3000 | Ambient temperature (°C)<br>24 |
|-------------------------------|-------------------------------------|--------------------------------|

**Pumps & water**

|                                           |                                           |                                      |                                    |
|-------------------------------------------|-------------------------------------------|--------------------------------------|------------------------------------|
| Fire pump rating (0 = auto) (gpm)<br>auto | Pump rated pressure (0 = auto) (bar)<br>0 | Number of duty pumps (0 = auto)<br>0 | Water storage (0 = auto) (m³)<br>0 |
|-------------------------------------------|-------------------------------------------|--------------------------------------|------------------------------------|

▶ Run simulation with my data  
 ✓ Recalculate  
 ↺ Reset to reference design  
 ↓ Export JSON  
 ↑ Import JSON

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Design Data: NFPA calculation steps, editable values and compliance checks

Shows every design calculation step (NFPA 13 density/area, K-factor, Hazen–Williams, pump selection, NFPA 2001 agent quantity, NFPA 11 foam...) in SI and US units. Change any value in the editor, press **Apply** or **Apply & run**, and the simulation runs on your design. The compliance list shows which NFPA rules pass or fail. **Reset** returns to the reference design.

**Training**  
Interactive 3D training scenes based on the course videos: explore every component, then perform the procedure step by step on live physics. Results are saved to the Classroom.

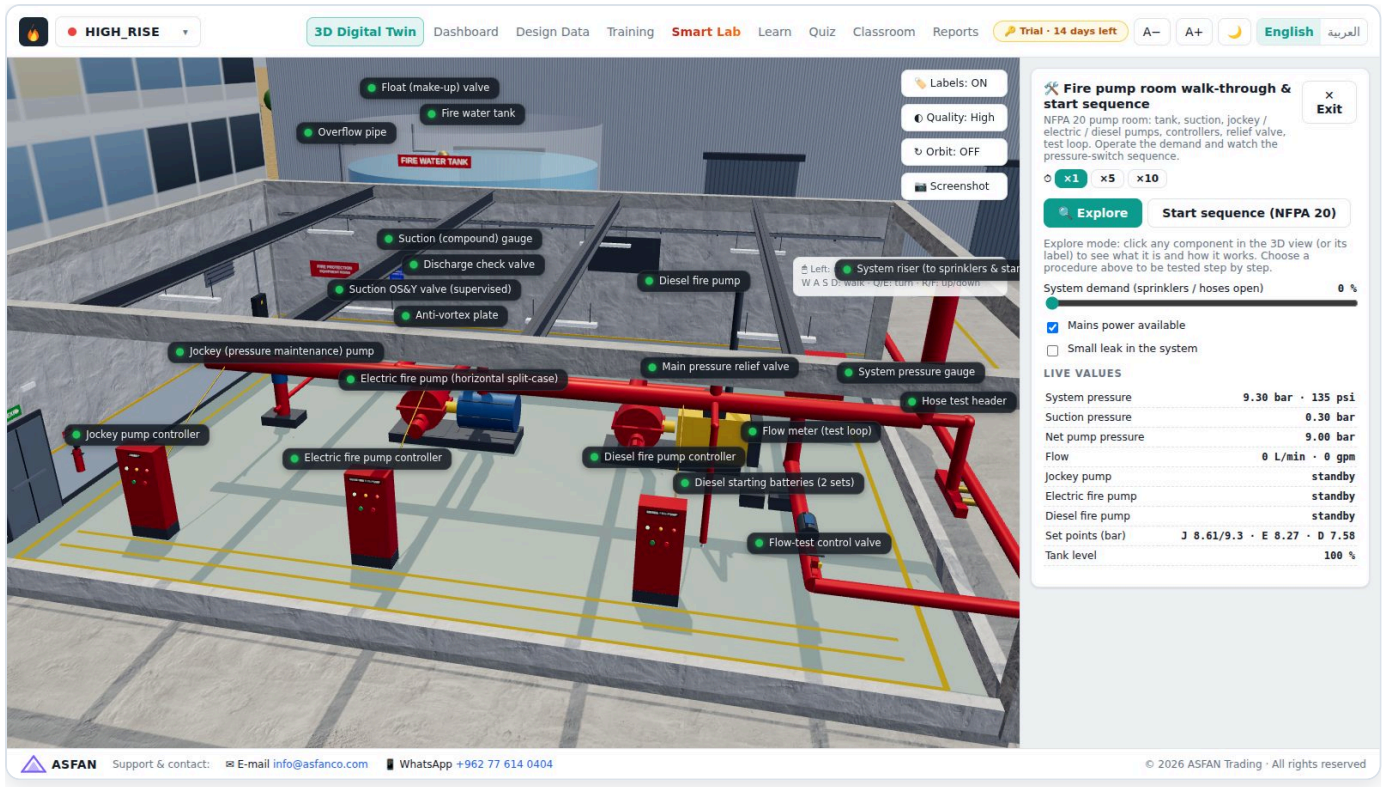
**New: Smart Systems Lab**  
Addressable panel, cause & effect, BMS integration, commissioning, predictive maintenance, incident command and certificates. [Open the Smart Lab →](#)

- Fire pump room walk-through & start sequence**  
NFPA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence. [▶ Start](#) [▶ Source video](#)
- Annual fire pump flow test (NFPA 25)**  
Run the pump at churn, 100 % and 150 % of rated flow through the flow-meter loop, record net pressure and compare with the acceptance curve. [▶ Start](#) [▶ Source video](#)
- Dry-pipe valve: trip test & reset**  
Air holds the clapper closed; opening the inspector's test lets air escape, the valve trips and water must reach the test outlet within the NFPA 13 limit. Then restore the system step by step. [▶ Start](#) [▶ Source video](#)
- Floor control valve assembly: alarm & tamper tests**  
Zone control valve with tamper switch, waterflow switch with retard, gauge, test-and-drain with sight glass and PRV. Run the waterflow alarm test and the valve supervisory test. [▶ Start](#) [▶ Source video](#)
- Portable extinguishers: placement & PASS**  
Place 2-A:10-B:C extinguishers on a light-hazard office floor so every point is within 22.9 m (75 ft) travel distance (NFPA 10), then fight a fire using P-A-S-S. [▶ Start](#) [▶ Source video](#)
- Hose drill: landing valve & fire hose cabinet**  
Two-person team: open the cabinet, run out the 65 mm hose without kinks, couple and tug-test, fit the branch nozzle, open the landing valve slowly and attack the fire at its base. [▶ Start](#) [▶ Source video](#)
- Sprinkler types & response race**  
Compare pendent, upright, sidewall, concealed, recessed, ESFR and open heads, then light a  $t^3$  fire under six bulbs of different temperature rating and RTI and watch which operates first. [▶ Start](#) [▶ Source video](#)
- Stairwell pressurization (NFPA 92)**  
Keep smoke out of the escape stair: set the supply fan so  $\Delta P \geq 12.5$  Pa, keep door-opening force  $\leq 133$  N, and see what happens when doors are opened.  $Q = 0.839 A \sqrt{\Delta P}$ . [▶ Start](#) [▶ Source video](#)
- FM-200 room: design calculator & discharge sequence**  
Size the agent live ( $W = V/S \times C/(100-C)$ ) with NOAEL/LOAEL warnings, then run the release sequence: cross-zoned detection, 30 s pre-discharge, HVAC shutdown, 10 s discharge and hold. [▶ Start](#) [▶ Source video](#)
- Installation mode: detectors & sprinklers**  
Lay out smoke detectors ( $0.7 \times S$  rule) and light-hazard pendent sprinklers on a real office ceiling with a deep beam, supply diffusers, an exposed duct and a column — live coverage, NFPA 72 / NFPA 13 obstruction rules, and an installer's layout to correct. [▶ Start](#) [▶ Source video](#)

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The Training page with the ten interactive 3D scenes

| Scene                  | What you practise                                                                  |
|------------------------|------------------------------------------------------------------------------------|
| Fire pump room         | NFPA 20 components and the jockey → electric → diesel start sequence, manual stop. |
| Annual pump flow test  | Churn, 100 % and 150 % flow, acceptance curve (NFPA 25).                           |
| Dry-pipe valve         | Full trip test with water-delivery time and the reset procedure.                   |
| Floor control valve    | Waterflow alarm test (retard) and tamper / supervisory test.                       |
| Portable extinguishers | Placement to NFPA 10 travel distance and the P-A-S-S technique.                    |
| Hose drill             | Landing valve, couplings and two-person team operation.                            |
| Sprinkler types        | Pendent, upright, sidewall, concealed, ESFR... and a response race.                |
| Stair pressurization   | NFPA 92 pressure difference, door-opening force, relief damper.                    |
| FM-200 room            | NFPA 2001 calculator and the cross-zoned discharge sequence.                       |
| 3D installation mode   | Placing detectors and sprinklers on a real ceiling with obstruction rules.         |



A training scene (fire pump room) with the procedure panel on the right

## How a scene works

- 1 **Explore:** click any component (or its label) to read what it is and how it works.
- 2 Choose a **procedure**. The steps appear on the right; the current step is highlighted.
- 3 Perform each step by clicking components, moving sliders or answering questions. Wrong actions count as mistakes.
- 4 Use ×1 / ×5 / ×10 to speed up waiting times. At the end you get a score; results are saved to the Classroom.

HIGH\_RISE

3D Digital Twin
Dashboard
Design Data
Training
Smart Lab
Learn
Quiz
Classroom
Reports
Trial - 14 days left
A-
A+
English
العربية

Labels: ON  
Quality: High  
Orbit: OFF  
Screenshot

Exit

**Fire pump room walk-through & start sequence**  
NFA 20 pump room: tank, suction, jockey / electric / diesel pumps, controllers, relief valve, test loop. Operate the demand and watch the pressure-switch sequence.

x1
x5
x10

Explore
Start sequence (NFA 20)

1. Check that the suction supply is open: click the supervised suction OS&Y valve.
2. Identify the jockey (pressure-maintenance) pump.
3. Open a small demand ( $\approx 2\%$ ). The jockey must start at 8.61 bar.
4. Increase the demand to  $\approx 60\%$  (sprinklers operating). The electric fire pump starts at 8.27 bar.
5. Close the demand to 0 %. Does the electric fire pump stop by itself?
6. With the demand closed, stop the electric pump: click its controller (STOP).
7. Simulate a mains power failure: switch off "Mains power available".
8. Open the demand again ( $\approx 60\%$ ). The electric pump cannot start; the diesel cranks at 7.58 bar.
9. Close the demand, restore mains power, then stop the diesel pump at its controller.

System demand (sprinklers / hoses open) 0 %

☒ Mains power available  
☐ Small leak in the system

LIVE VALUES  
System pressure 9.30 bar · 135 psi

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Procedure mode: current step, live values and the event log

## 10 Smart Systems Lab (advanced)

The **Smart Lab** tab trains engineers on smart (addressable) fire-alarm and life-safety systems. All ten modules share one live system: a fault planted in the panel appears in the building, the BMS, the monitoring centre and the commissioning records. A ✓ in the left menu marks completed modules.

The screenshot displays the 'Smart Systems Lab' interface. On the left, a sidebar lists ten modules, with the first one, 'Addressable Fire Alarm Panel Lab', marked as completed with a green checkmark. The main area is titled 'Addressable Fire Alarm Panel Lab' and includes a description: 'Operate a real-style addressable FACP connected to a 3-storey building; test every device in the field, cut and short the SLC loop to see Class A and isolators at work, program addresses with auto-learn, and diagnose hidden faults like a service engineer.' Below this, there are tabs for 'Live panel & building', 'Programming', 'Guided exercises', 'Inspector challenge', and 'Devices: install & wiring'. The 'Live panel & building' tab is active, showing a digital representation of the 'ASFAN FX-3000' fire alarm control panel. The panel's LCD displays 'SYSTEM NORMAL', 'Loop 1: 61 devices OK Class A', 'AC ON BATT 27.3V', and 'ACCESS LEVEL 1'. It also shows a time of '08:55:51' and a voltage of 'PSU 230 VAC - 27.3 V'. The panel has several buttons: 'ACKNOWLEDGE', 'ALARM SILENCE', 'RESOUND', 'SYSTEM RESET', 'DRILL', 'LAMP TEST', and 'BUZZER'. To the right of the panel is a floor plan diagram of a 3-storey building with various rooms and a network of red lines representing the SLC loop. A toolbar above the floor plan allows users to 'Select / test', 'Cut cable', 'Short', 'Repair', and 'Add device'. A legend below the floor plan identifies symbols for SLC loop, NAC circuit, Isolator, Call point, Module, and Detector. At the bottom of the interface, there is a footer with 'ASFAN' logo, contact information, and a copyright notice for 2026 ASFAN Trading.

Module 1 – addressable fire alarm panel with the building plans and SLC loop

### 1 · Addressable panel lab

A realistic panel (LCD, LEDs, **ACKNOWLEDGE**, **ALARM SILENCE**, **SYSTEM RESET**, **DRILL**, lamp test and the access-level key switch). Click a device on the plan to test it (smoke, heat, call point, waterflow, valve). Use the toolbar to **cut** or **short** the loop, compare Class A and B, see isolators work, add devices and run **auto-learn** in the Programming tab. **Guided exercises** are detected automatically; the **Inspector challenge** hides faults for you to find. Silence and reset need access level 2 — turn the key.





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Dashboard

Design Data

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Smart Lab

Learn

Quiz

Classroom

Reports

Trial - 14 days left

A-

A+

English

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ADVANCED

Smart Systems Lab

Install, program, integrate, commission and maintain an addressable fire-alarm system.

1

Addressable Fire Alarm Panel Lab

SLC loop, isolators, faults, progr...

2

Cause & Effect Matrix

Program the smart logic & run fl...

3

BMS & Building Integration

BACnet/Modbus, lift, voice evac...

4

Installation Mode (3D)

Detector & sprinkler layout rules

5

Commissioning & Acceptance Testing

Acceptance tests & record of co...

6

Predictive Maintenance

Condition monitoring, ITM & wor...

7

Incident Commander

Decisions under pressure, casca...

8

Modern Detection & Suppression Technologies

ASD, beam, flame, LHD, mist, pr...

9

Engineering Tools

Hydraulics, batteries, voltage dr...

10

Learning Paths & Certificates

Levels, final exam, certificates

✓

Commissioning & Acceptance Testing

Take the installed system through acceptance exactly as a commissioning engineer does: pre-functional inspection and cable tests, a device-by-device functional test on the live panel, audibility and visibility survey, battery and power tests, the sprinkler hydrostatic test — and issue the NFPA 72 Record of Completion.

NFPA 72 §14.4 · T.14.4.3.2

NFPA 72 §7.5.6

NFPA 72 §18.4 · §18.5

NFPA 72 §10.6

NFPA 13 §29.2.1

1 · Pre-functional

2 · Device functional test

3 · Audibility & visibility

4 · Battery & power

5 · Hydrostatic test

6 · Record of Completion

Checklist items

0/14

Flagged entries

0

Installation defects recorded

0

Loop class

A

Programmed devices

61

PRE-FUNCTIONAL INSPECTION & CABLE TESTS

NFPA 72 §14.4 · T.14.4.3.2

| Item / acceptance criterion                                                                                                                    | Measured                                        | Verdict                         |
|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------|
| <div>○</div> <div>Addressable devices, panel and surge protectors disconnected before insulation test</div> <div>Mfr. / NFPA 72 §14.2.10</div> | <div>☐</div> <div>Verified on site</div>        |                                 |
| <div>○</div> <div>SLC loop cable insulation resistance (conductors → earth)</div> <div>≥ 1 MΩ @ 500 V DC · NFPA 72 T.14.4.3.2 (21)</div>       | <div>—</div> <div>MΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>NAC1 cable insulation resistance</div> <div>≥ 1 MΩ @ 500 V DC · NFPA 72 T.14.4.3.2 (21)</div>                                | <div>—</div> <div>MΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>NAC2 cable insulation resistance</div> <div>≥ 1 MΩ @ 500 V DC · NFPA 72 T.14.4.3.2 (21)</div>                                | <div>—</div> <div>MΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>SLC loop resistance, panel OUT → IN (continuity)</div> <div>≤ 40 Ω · NFPA 72 T.14.4.3.2 (21)</div>                           | <div>—</div> <div>Ω</div> <div>Measure</div>    | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>Earth-fault monitor: loop resistance to earth (panel reading)</div> <div>≤ 50 kΩ · NFPA 72 §12.6</div>                       | <div>—</div> <div>kΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>NAC1 end-of-line resistor</div> <div>4.7 kΩ ± 5 % · NFPA 72 §12.6</div>                                                      | <div>—</div> <div>kΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>NAC2 end-of-line resistor</div> <div>4.7 kΩ ± 5 % · NFPA 72 §12.6</div>                                                      | <div>—</div> <div>kΩ</div> <div>Measure</div>   | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>Loop scan: devices answering = devices programmed, no double address</div> <div>= programmed · NFPA 72 §14.4.1</div>         | <div>—</div> <div>dev.</div> <div>Measure</div> | <div>Pass</div> <div>Fail</div> |
| <div>○</div> <div>Max. smoke-detector spacing, smooth ceiling (corridor)</div> <div>≤ 18.3 m · NFPA 72 §15.2.5.5</div>                         | <div>—</div> <div>m</div> <div>Measure</div>    | <div>Pass</div> <div>Fail</div> |

TEST INSTRUMENT

Why disconnect first?

A 500 V DC insulation test destroys the electronics of addressable detectors, modules and the panel. Megger the cables with devices removed from their bases and conductors lifted at the panel.

ASFAN

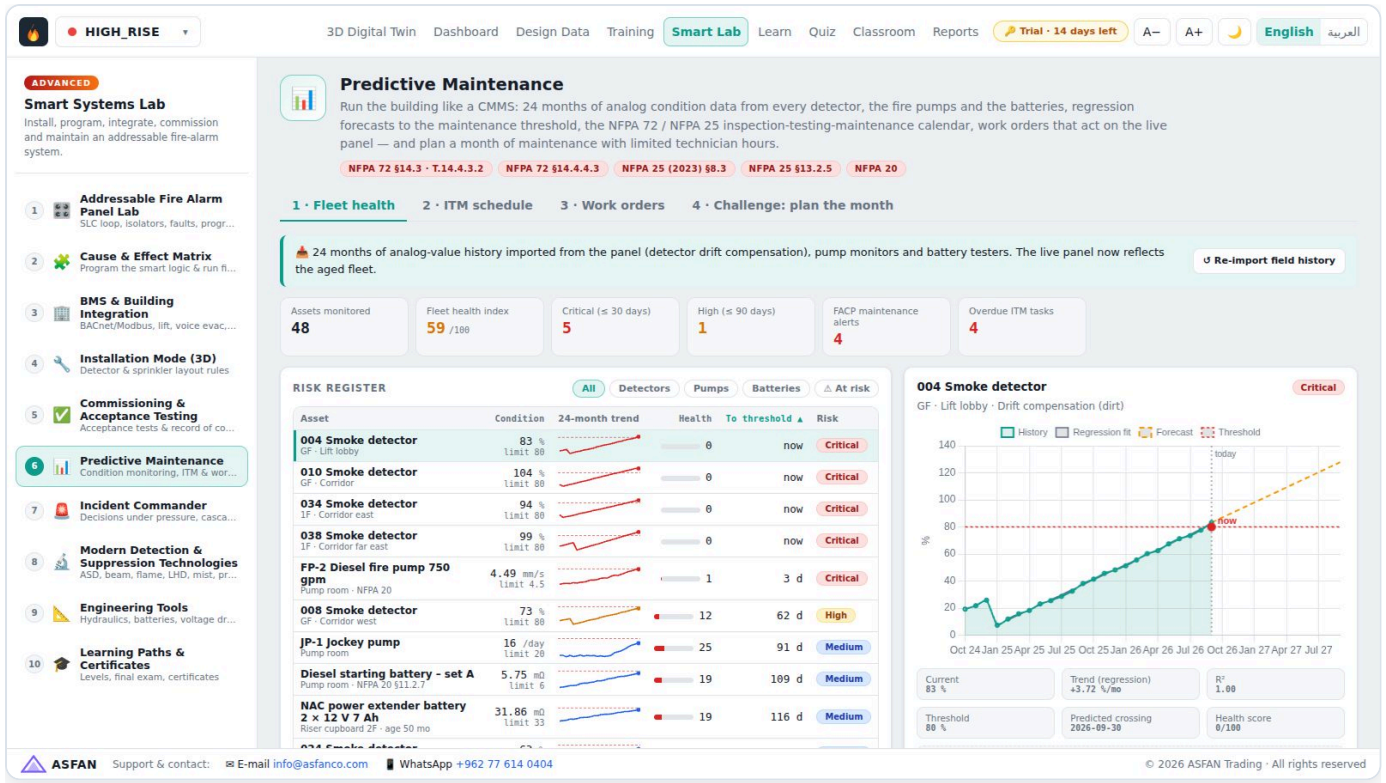
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## Module 5 – commissioning: device tests, audibility, batteries, hydrostatic test, record

### 6 · Predictive maintenance

Condition data for detectors, pumps and batteries with trend forecasts, the NFPA 72 / NFPA 25 inspection-testing-maintenance calendar, work orders that really clean detectors on the panel, and a monthly planning challenge.



## Module 6 – fleet health, risk register and forecast

### 7 · Incident commander

Timed decisions in fires with cascading failures (night server-room fire, office-hours kitchen fire). Afterwards **Replay & debrief** shows your timeline, the best actions and their code basis.

### 8 · Modern detection & suppression

Simulators for aspirating smoke detection (VESDA), beam detectors, UV/IR flame detectors, linear heat cable, water mist and pre-action systems (non-, single- and double-interlock), plus a technology-selection challenge.



## 10 · Learning paths & certificates

Enter your name, follow a path (Technician, Engineer, Supervisor, Designer), complete the required modules, pass the final exam ( $\geq 80\%$ ) and print a certificate with a verification code. Supervisors see the instructor dashboard with each trainee's progress.



HIGH\_RISE

3D Digital Twin

Dashboard

Design Data

Training

Smart Lab

Learn

Quiz

Classroom

Reports

Trial - 14 days left

A-

A+

English

العربية

ADVANCED

**Smart Systems Lab**  
Install, program, integrate, commission and maintain an addressable fire-alarm system.

1

**Addressable Fire Alarm Panel Lab**  
SLC loop, isolators, faults, progr...

2

**Cause & Effect Matrix**  
Program the smart logic & run fi...

3

**BMS & Building Integration**  
BACnet/Modbus, lift, voice evac...

4

**Installation Mode (3D)**  
Detector & sprinkler layout rules

5

**Commissioning & Acceptance Testing**  
Acceptance tests & record of co...

6

**Predictive Maintenance**  
Condition monitoring, ITM & wor...

7

**Incident Commander**  
Decisions under pressure, casca...

8

**Modern Detection & Suppression Technologies**  
ASD, beam, flame, LHD, mist, pr...

9

**Engineering Tools**  
Hydraulics, batteries, voltage dr...

10

**Learning Paths & Certificates**  
Levels, final exam, certificates

**Learning Paths & Certificates**  
Four professional paths — Technician, Engineer, Supervisor, Designer. Complete the required labs, pass the final exam ( $\geq 80\%$ ) and print a certificate with a verification code that any employer can check offline in this program.  
**NFPA 72 §10.5 Personnel qualifications** **NFPA 25 §4.4** **NICET Level I-IV (reference)**

Learning paths

Final exam

Certificates & verification

Instructor dashboard

Trainee name (printed on the certificate)

Group / company

Full name

Save

**Fire Alarm Technician**  
Install, test and maintain addressable systems.

0%

Panel lab

0% / 60%

Installation

0% / 60%

Commissioning

0% / 60%

Predictive maintenance

0% / 60%

Final exam

Complete the labs first

**Fire Safety Supervisor**  
Operate the building in emergencies and manage ITM & impairments.

0%

Cause & effect

0% / 70%

BMS integration

0% / 80%

**Smart Systems Engineer**  
Program, integrate and commission complete smart fire systems.

0%

Panel lab

0% / 80%

Cause & effect

0% / 80%

BMS integration

0% / 60%

Installation

0% / 80%

Commissioning

0% / 70%

Engineering tools

0% / 60%

Final exam

Complete the labs first

**Fire Protection Designer**  
Design layouts, hydraulics, power and logic to code.

0%

Cause & effect

0% / 90%

Installation

0% / 90%

ASFAN

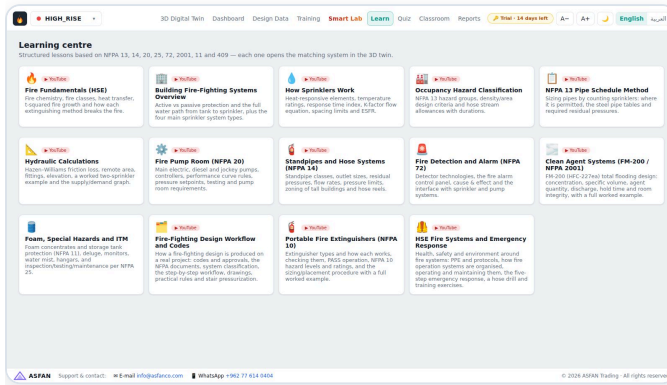
Support & contact: E-mail [info@asfanco.com](mailto:info@asfanco.com) WhatsApp +962 77 614 0404

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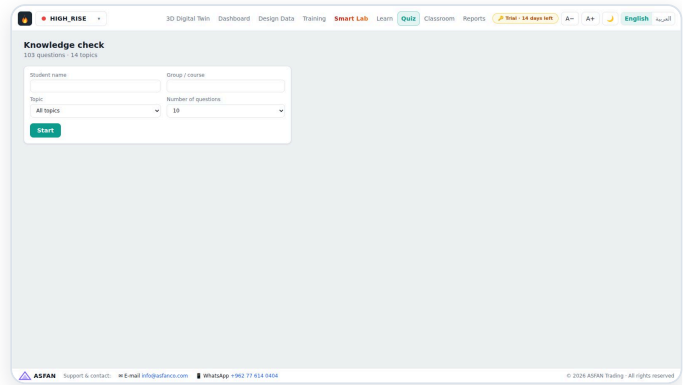
### Module 10 – learning paths, final exam and certificates

Fire Protection Digital Twin · User Manual

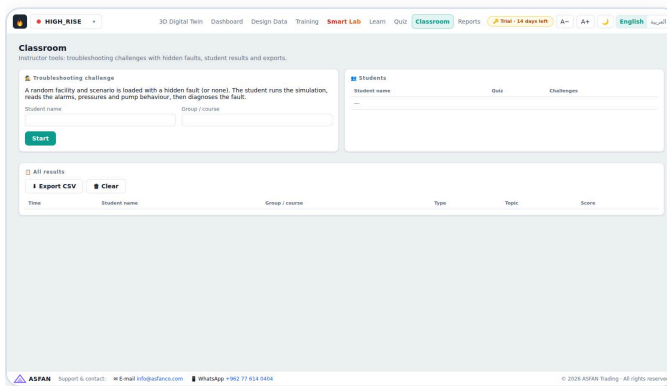
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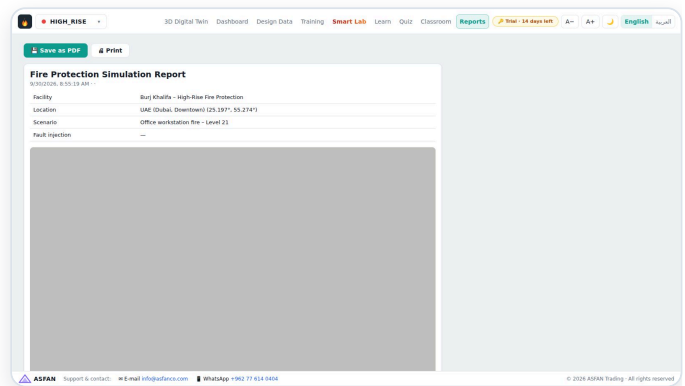
Learn – bilingual lessons with source videos



Quiz – questions with explanations



Classroom – challenges and results



Reports – printable simulation report

- **Learn:** illustrated lessons (fire basics, sprinklers, hazards, hydraulics, pumps, standpipes, detection & alarm, clean agents, foam, ITM) with links to the source videos.
- **Quiz:** choose a topic and number of questions; each answer shows an explanation.
- **Classroom:** enter the student name and group, start a **troubleshooting challenge** (hidden fault), and see results. A supervisor license can export and clear all results (CSV).
- **Reports:** **Save as PDF** or **Print** a full simulation report with screenshot, calculations and event log.

| Problem                                                                                                 | Solution                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| "Windows protected your PC" when installing                                                             | Click <b>More info</b> → <b>Run anyway</b> . The program is safe; the message appears because it is not signed with a paid certificate.             |
| 3D view is black, slow or flickering                                                                    | Switch <b>Quality</b> to <b>Low</b> ; update the graphics driver; on laptops choose the dedicated GPU for the program in Windows Graphics settings. |
| "License is bound to another computer"                                                                  | Send your current Machine ID to ASFAN to receive a key for this computer.                                                                           |
| "System clock rollback detected"                                                                        | Set the correct date and time in Windows and restart the program.                                                                                   |
| A tab or module shows  | It is not included in your package. Contact ASFAN to upgrade.                                                                                       |
| Text too small / large                                                                                  | Use <b>A-</b> / <b>A+</b> in the top bar.                                                                                                           |
| Results disappeared                                                                                     | Results are stored per Windows user; log in with the same Windows account.                                                                          |

**ASFAN Trading**

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When you contact support, please send: your name / organisation, your Machine ID (license window), the program version (Help → About) and a screenshot of the problem.